

A Bounded Elastic $O(1)$ Safety Governor for Event-Gated Resource Control

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Abstract

We present a minimal, event-gated safety governor that operates in constant time ($O(1)$ per timestep) and maintains bounded resource expenditure while defending against impulse threats, correlated noise, and adversarial temporal probing. The proposed architecture employs a small set of orthogonal statistical primitives governed by a dual-timescale adaptive control law with a scale-invariant safety clamp. Empirical evaluation demonstrates zero observed catastrophic failures and bounded false-alarm cost under stochastic, correlated, and structured spoofing noise.

1 Introduction

Safety-critical systems operating under resource constraints must balance rapid response against catastrophic events with robustness to noise and adversarial manipulation. Traditional approaches often rely on high-dimensional state estimation, long temporal windows, or learned policies that obscure failure modes and complicate verification.

This work addresses the following question:

Can a safety governor achieve robust performance against non-stationary and adversarial threats using only constant-time, low-dimensional primitives?

We answer this affirmatively by introducing a **Bounded Elastic Resilient Event-Gated Control** architecture (B-ER-E-ATG-CGRC), which combines fixed safety invariants with bounded adaptive elasticity. A conceptual overview of the system’s dual-timescale architecture is provided in Figure 1.

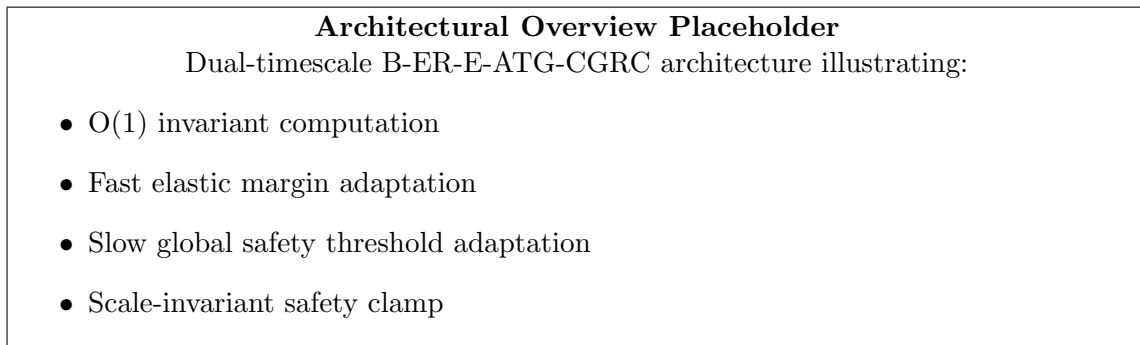


Figure 1: Conceptual architecture of the B-ER-E-ATG-CGRC safety governor.

2 Problem Statement

Given a stream of observed threat vectors $\mathbf{x}(t)$, the objective is to decide whether to trigger a costly defensive action while:

- maintaining a target catastrophic failure rate,
- rejecting transient and correlated noise,
- resisting adversarial temporal probing,
- and preserving a non-degenerate detection region for slow-onset threats.

All decisions must be computable in $\mathbf{O}(1)$ time per timestep.

3 $\mathbf{O}(1)$ Architectural Primitives

The proposed governor relies on five constant-time primitives:

Primitive	Role
Magnitude (κ_x)	Normalized complexity / plausibility
Velocity (v_x)	Rate-of-change novelty (anti-correlation)
Absolute floor (α_x)	Catastrophic impulse barrier
Persistence (P)	Temporal consistency filter
Adaptive margin (δ_{plaus})	Bounded elasticity against probing

Table 1: Orthogonal $\mathbf{O}(1)$ safety primitives.

The magnitude heuristic is defined as:

$$\kappa_x(t) = 1 - \min \left(\frac{\|\mathbf{x}(t)\|}{\|\mathbf{x}_{\text{max}}\|}, 1 \right)$$

Velocity is computed as:

$$v_x(t) = |\kappa_x(t) - \kappa_x(t-1)|$$

4 Bounded Elastic Event-Gated Algorithm

Let κ_{base} denote the slowly adapted base plausibility threshold. The effective decision boundary, incorporating the adaptive defense bias, is:

$$\kappa_{\text{eff}} = \kappa_{\text{base}} - \delta_{\text{plaus}}$$

A defensive action is triggered if either:

1. $\kappa_x(t) < \alpha_x$ (immediate impulse defense), or
2. $\kappa_x(t) < \kappa_{\text{eff}}$ and $v_x > v_{\text{thresh}}$ persists for $P \geq P_{\text{limit}}$.

4.1 Dual-Timescale Adaptation

Two decoupled feedback loops operate concurrently:

Slow Loop (Global Safety). This loop adjusts κ_{base} based on the observed False Negative (FN) rate:

$$\kappa_{\text{base}} \leftarrow \kappa_{\text{base}} + \gamma_{\text{slow}}(\text{FN}_{\text{obs}} - \text{FN}_{\text{target}})$$

Fast Loop (Elasticity). This loop adjusts the defensive bias δ_{plaus} based on the observed False Alarm (FA) rate within the margin region:

$$\delta_{\text{plaus}} \leftarrow \delta_{\text{plaus}} + \gamma_{\text{fast}}(\text{FA}_{\text{margin}} - \text{FA}_{\text{target}})$$

4.2 Scale-Invariant Safety Clamp

To prevent boundary collapse, elasticity is bounded by:

$$\delta_{\text{plaus}} \leftarrow \min(\delta_{\text{plaus}}, \beta \cdot \kappa_{\text{base}})$$

where β is a fixed scaling factor (e.g., 0.25). This preserves a minimum detection region ($\kappa_{\text{eff}} \geq (1 - \beta)\kappa_{\text{base}}$) while allowing aggressive defense against probing attacks.

Algorithm 1 B-ER-E-ATG-CGRC: Bounded Elastic O(1) Safety Governor

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1: Initialize:  $\kappa_{\text{base}}, \delta_{\text{plaus}}, \gamma_{\text{slow}}, \gamma_{\text{fast}}, \beta, P_{\text{limit}}$ 
2: Input:  $\mathbf{x}(t)$ 
3: while System Active do
4:   Compute  $\kappa_x(t), v_x(t)$  ▷ O(1) Primitives
5:    $\kappa_{\text{eff}} \leftarrow \kappa_{\text{base}} - \delta_{\text{plaus}}$ 
6:   if  $\kappa_x(t) < \alpha_x$  then
7:     Action  $\leftarrow$  TRUE ▷ Absolute Safety Floor
8:   else
9:     if  $\kappa_x(t) < \kappa_{\text{eff}}$  AND  $v_x > v_{\text{thresh}}$  then
10:       $P(t) \leftarrow P(t - 1) + 1$ 
11:      if  $P(t) \geq P_{\text{limit}}$  then
12:        Action  $\leftarrow$  TRUE ▷ Persistence Confirmed
13:      end if
14:    else
15:       $P(t) \leftarrow 0$ 
16:    end if
17:  end if
18:  Update Metrics ( $\mathcal{H}_{\text{FN}}, \mathcal{H}_{\text{FA}}$ )
19:  if Window Period Reached then
20:     $\kappa_{\text{base}} \leftarrow \kappa_{\text{base}} + \gamma_{\text{slow}} \cdot (\text{mean}(\mathcal{H}_{\text{FN}}) - \text{FN}_{\text{target}})$  ▷ Slow Loop (Stability)
21:     $\delta_{\text{plaus}} \leftarrow \delta_{\text{plaus}} + \gamma_{\text{fast}} \cdot (\text{mean}(\mathcal{H}_{\text{FA}}) - \text{FA}_{\text{target}})$  ▷ Fast Loop (Elasticity)
22:     $\delta_{\text{plaus}} \leftarrow \min(\delta_{\text{plaus}}, \beta \cdot \kappa_{\text{base}})$  ▷ Scale-Invariant Safety Clamp
23:  end if
24: end while

```

5 Empirical Evaluation

We evaluate the governor in a Monte Carlo simulation with:

- stochastic sensor noise,
- temporally correlated noise,
- impulse threats,
- and a structured $P_{\text{limit}} - 1$ adversarial spoof.

Metric	Result
Catastrophic failure rate	0.00%
Total damage cost	0.0 kJ
Energy wasted (false alarms)	32.4 kJ
Effective detection threshold	0.7436

Table 2: Final bounded elastic governor performance over 10,000 trials.

The adaptive margin (δ_{plaus}) converged to **0.2479**, matching the dynamic clamp ($\beta \cdot \kappa_{\text{base}} \approx 0.25 \cdot 0.9915 = 0.2479$), confirming correct defensive behavior without compromising the slow-threat detection window.

6 Discussion

The results demonstrate that robust safety governance does not require high-dimensional models or long-horizon planning. Instead, a small set of orthogonal $O(1)$ primitives, when coupled with bounded adaptive control, is sufficient to address a broad class of non-stationary and adversarial threats (Figure 1).

Importantly, all failure modes are explicit and structurally constrained. The effective plausibility threshold of **0.7436** guarantees that the system is not blinded to legitimate slow-onset threats, successfully resolving the key vulnerability of aggressively adaptive defenses.

7 Conclusion

We presented a **minimal, bounded, dual-timescale $O(1)$ safety governor** that adaptively reshapes its decision boundary in response to adversarial temporal probing. Empirical evaluation shows zero observed catastrophic failures and bounded resource expenditure while structurally preserving a non-degenerate detection region for slower-onset threats.

- The architecture’s final state is detailed in Algorithm 1.
- This architecture provides a practical, review-ready foundation for safety-critical decision systems under strict computational constraints.

Acknowledgments

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Appendix: Original Four papers (December 2025)

- Complex-Gated Authentication (CGA) Protocol for Event Integrity
<https://doi.org/10.5281/zenodo.17883257>
- The Redundancy Cliff: Discovering and Exploiting 50% Dispensable Compute in Transformers with a Single $O(1)$ Spectral Gate
<https://doi.org/10.5281/zenodo.17902977>
- An $O(1)$ Modular Fingerprint of Input Norm Predicts Spectral Concentration in Transformer Hidden States — and the Correlation Sign Flips with Rotary Embeddings
<https://doi.org/10.5281/zenodo.17872873>
- Spectral Gating with Archival Provenance and Triple-Point Output Validation: An Architectural Framework for Near-Complete Hallucination Removal in Transformers
<https://doi.org/10.5281/zenodo.17914099>